

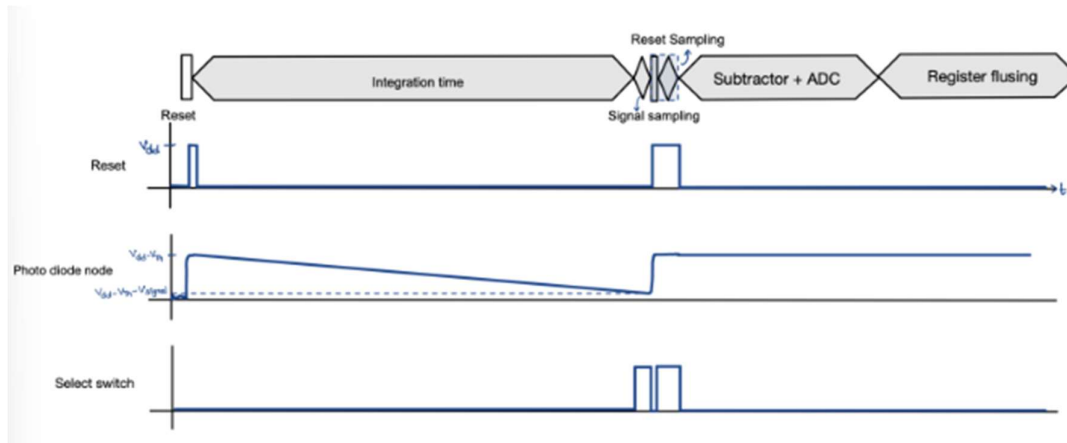
CMOS IMAGE SENSOR

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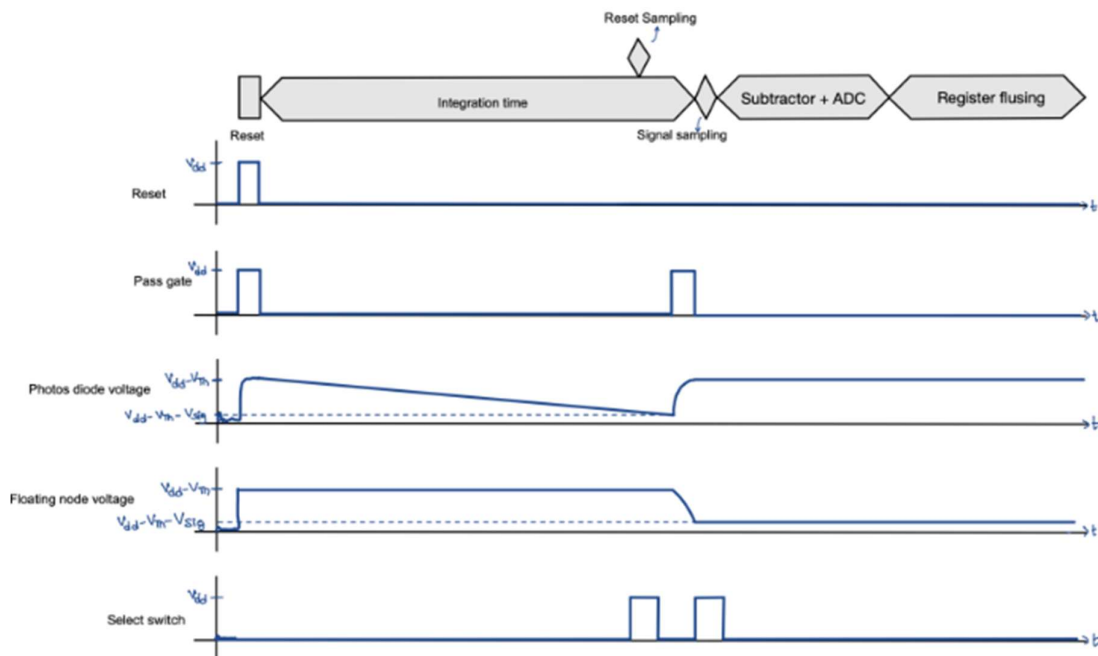
1. 3-Transistor Pixel

To simulate the generation of photo-current, I used a current source pumping current from the FD node to GND. I found the FD node capacitance by keeping the node floating and then pumping out a current of few nA from that node, then by observing the slope we calculate the Capacitance to be kept at FD (CFD).



$$T_{row\ read} = T_{Reset} + T_{Integration} + T_{Signal\ Sampling} + T_{Reset} + T_{Reset\ Sampling} + T_{Subtractor+ADC} + T_{Register\ fusing}$$

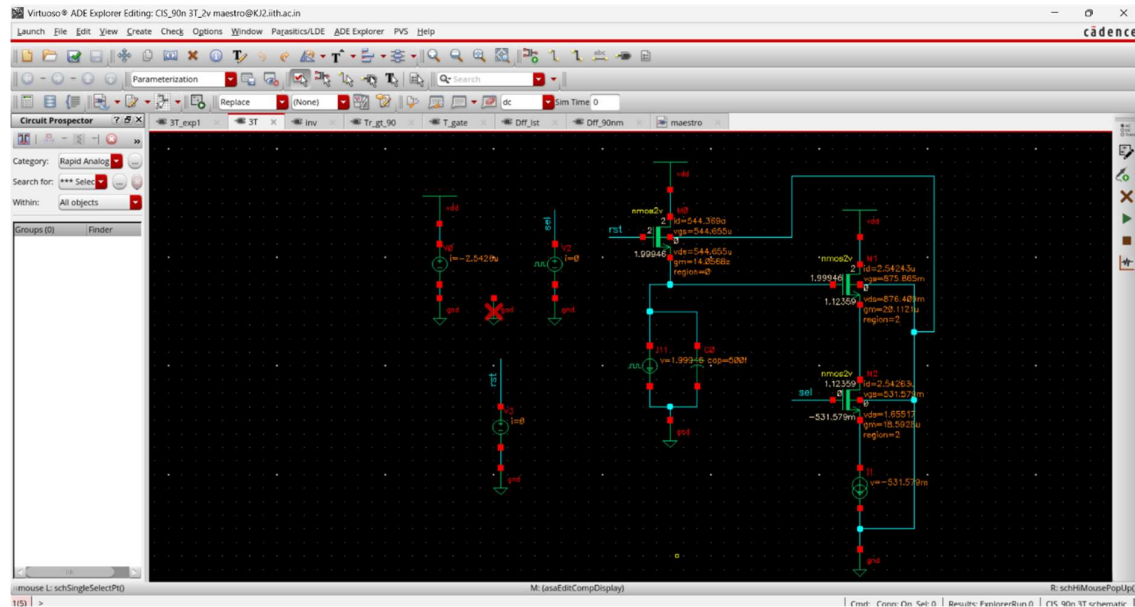
2. 4-Transistor Pixel



The above image shows the timing of a series of enables/switches of nodes that occur in making a reading on one row for the first image. Time to read one row counting from rest of photo diode:

$T_{row\ read} = T_{Reset} + T_{Integration} + T_{Signal\ Sampling} + T_{Subtractor+ADC} + T_{Register\ flushing}$

3. Modelling of 3T pixel



A 3T (Three-Transistor) Active Pixel Sensor is one of the earliest CMOS image sensor architectures used for converting incident light into electrical signals. The pixel consists of a photodiode and three transistors that perform reset, buffering, and readout operations.

The 3T pixel consists of:

1. **Photodiode (PD)** – Converts incident photons into charge.
2. **Reset Transistor (RST)** – Resets the photodiode node.
3. **Source Follower (SF)** – Buffers the pixel voltage.
4. **Row Select (RS)** – Connects the pixel output to the column readout circuitry.

Working Mechanism of a 3T Pixel

1. Reset Phase

- Reset transistor (RST) is turned ON.

- Photodiode node is charged to the reset voltage.
- Any previously accumulated charge is removed.

Result

$$V_{PD} = V_{RST}$$

The photodiode starts from a known initial condition.

2. Integration Phase

- Reset transistor is turned OFF.
- Incident light generates electron-hole pairs.
- Electrons accumulate in the photodiode.

Result

The photodiode voltage decreases as charge accumulates.

$$Q_{photo} \propto \text{Light Intensity}$$

3. Readout Phase

- Row Select transistor is turned ON.
- Source Follower buffers the photodiode voltage.
- Signal is sent to the column readout circuit.

Output

The output voltage is proportional to the amount of incident light.

Limitations of 3T Pixel

Reset Noise (kTC Noise)

$$V_n = \sqrt{\frac{kT}{C}}$$

This reset noise directly appears in the output signal.

Dark Current

- Leakage currents generate unwanted charge.
- Produces image degradation under low-light conditions.

Lower Dynamic Range

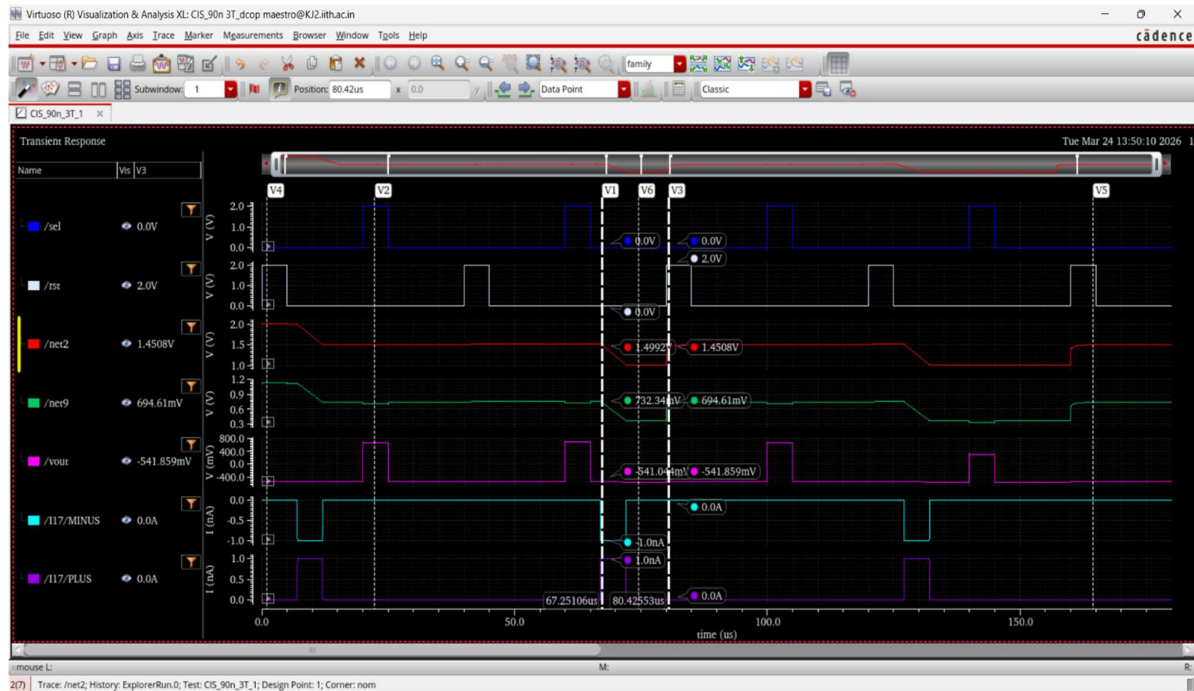
- Reduced ability to distinguish between very bright and very dark regions.

No Charge Transfer Mechanism

Unlike 4T pixels:

- Charge remains inside the photodiode.
- No separate floating diffusion node exists.

Simulation result:



Rst is done before the select so that it will have the highest possible value at the FD so that the photo diode current which is generated by the current source gets captured and thus can be seen at the output perfectly.

4. Modelling of 4T pixel

A 4T Active Pixel Sensor improves upon the 3T architecture by introducing:

- Transfer Gate (TX)
- Pinned Photodiode (PPD)

This allows charge-domain operation and enables Correlated Double Sampling (CDS), significantly reducing noise and improving image quality.

4T Pixel Structure

The 4T pixel consists of:

1. Pinned Photodiode (PPD)
2. Transfer Gate (TX)
3. Reset Transistor (RST)

4. Source Follower (SF)
5. Row Select (RS)

Working Mechanism of a 4T Pixel

1. Reset Phase

- Floating Diffusion (FD) node is reset.
- FD is charged to reset voltage.

2. Integration Phase

- Light falls on the PPD.
- Electrons accumulate in the pinned photodiode.

3. Charge Transfer Phase

- Transfer Gate (TX) is enabled.
- Charge moves from PPD to FD node.

Benefit

Efficient charge transfer improves sensitivity and reduces noise.

4. Readout Phase

- Source Follower converts charge into voltage.
- Row Select enables column readout.

5. Correlated Double Sampling (CDS)

Step 1

Sample the reset level.

Step 2

Sample the signal level after charge transfer.

Step 3

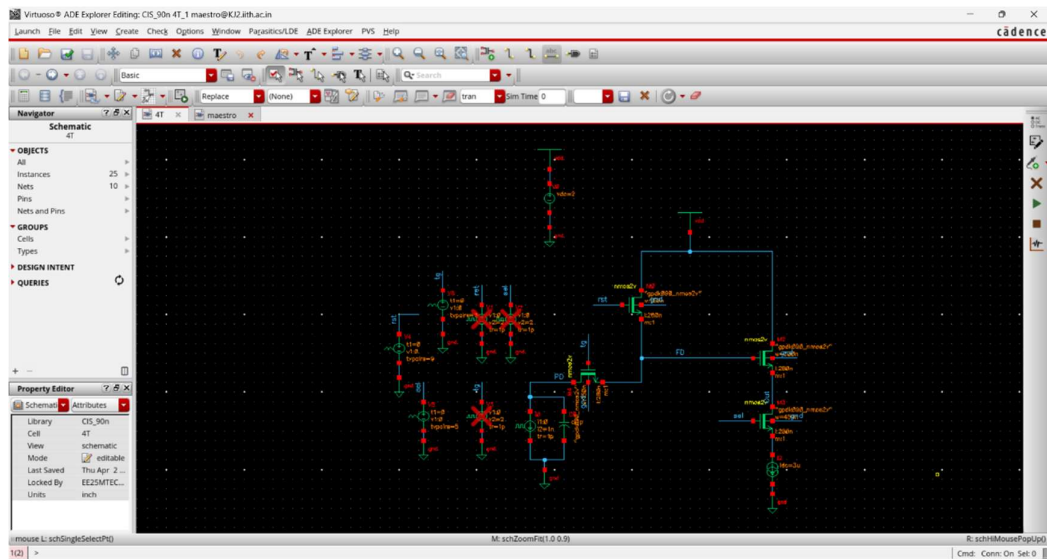
Subtract the two samples:

$$V_{signal} = V_{sample} - V_{reset}$$

Result

- Reset noise cancellation
- Reduced Fixed Pattern Noise (FPN)

Schematic



Simulation



Current Steering DAC (my motive is to generate a ramp)

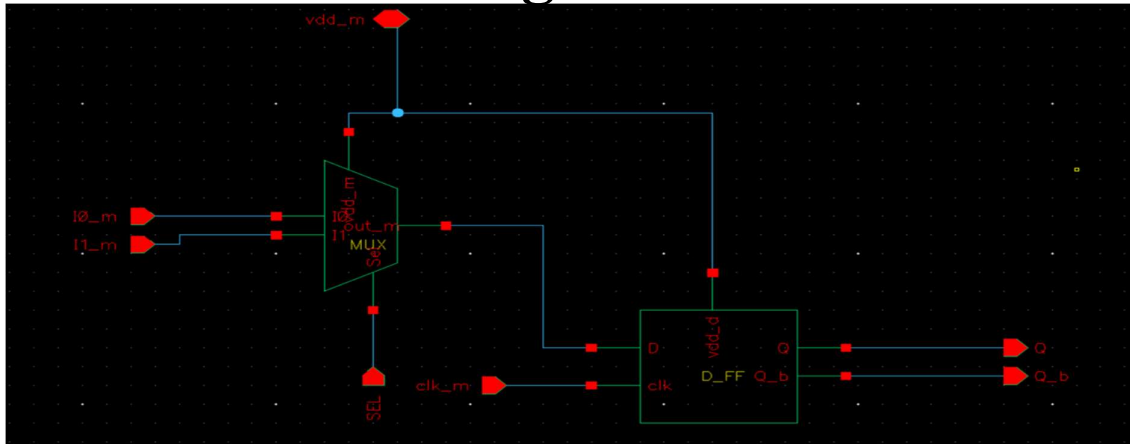


Fig:1 Dff with preset (symbol name in schematic – Dff_1)

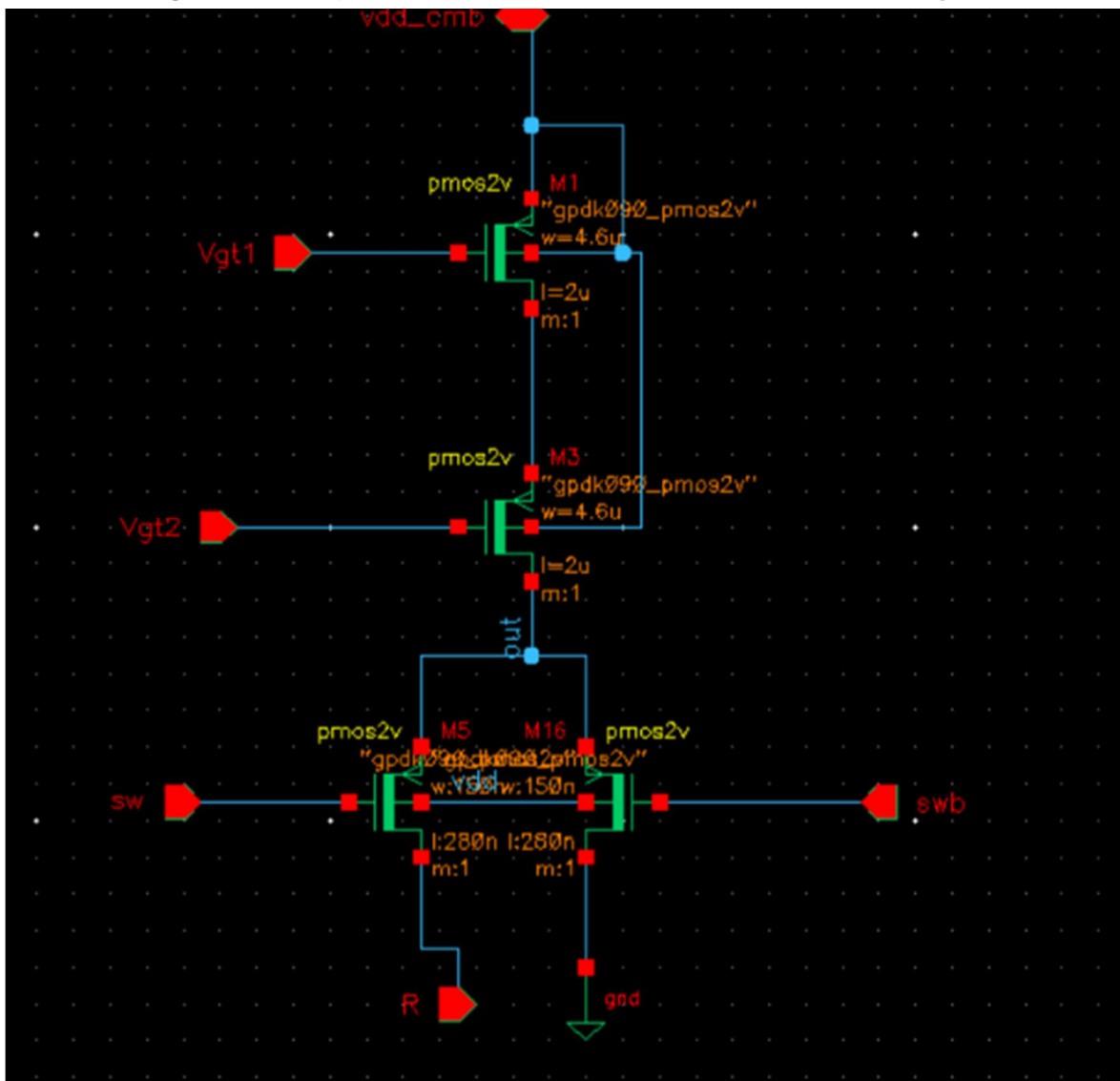


Fig:2 Individual Current branch(I've used 256 branches so that my DAC levels are from 256 to 0) ...symbol name in cad is **CURR_BRANCH**

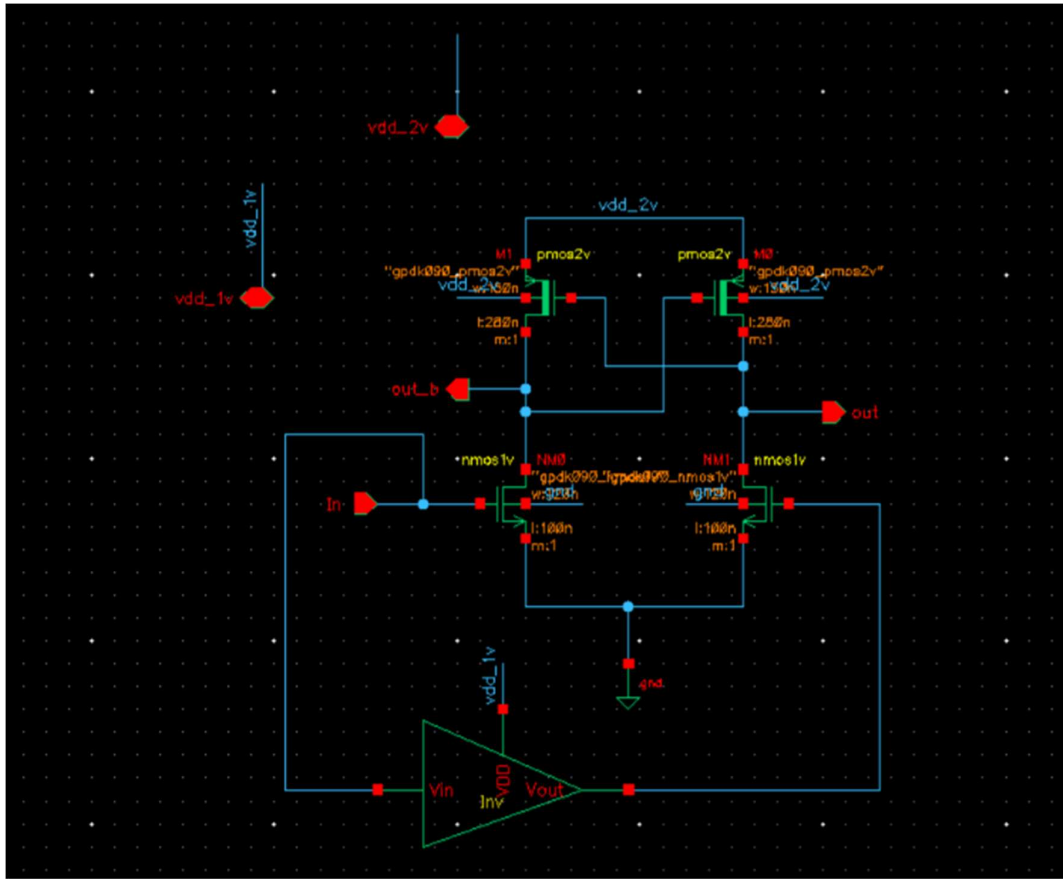


Fig:3 Level shifter (my Dff is **nmos1v** which operates in low voltage, but my output of DAC is a high voltage parameter.. so used a level shifter to shift my 1v to 2v (can also skip if both input and output signals are either low voltage or high voltage accordingly)
symbol name in cad – **lv1_shifter**

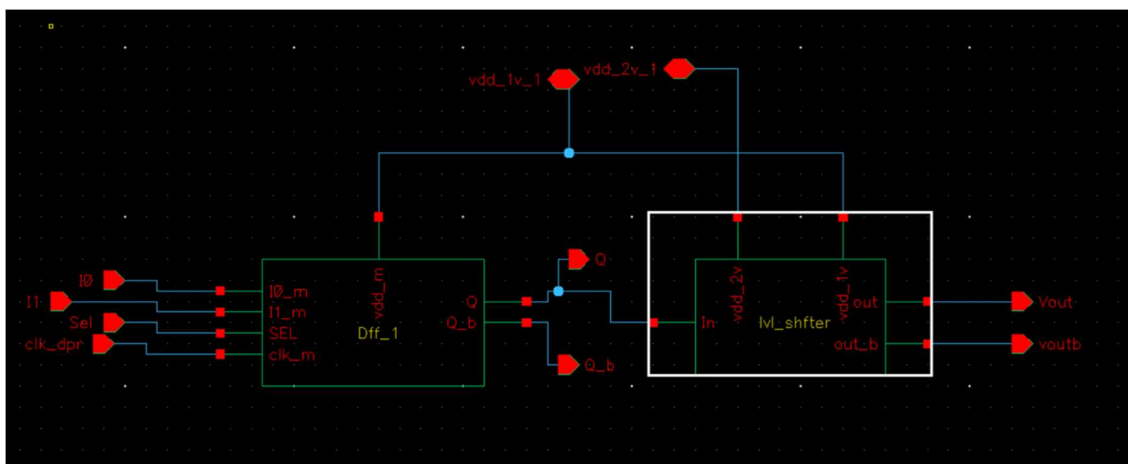


Fig:4 Here I have connected both **Dff with preset (Dff_1)** and **level shifter(lv1_shifter)**
whose input takes **0-1V** and output gives **0-2V**

Fig4's symbol is cad is **dff_pr_ls**(can be seen in the next fig)

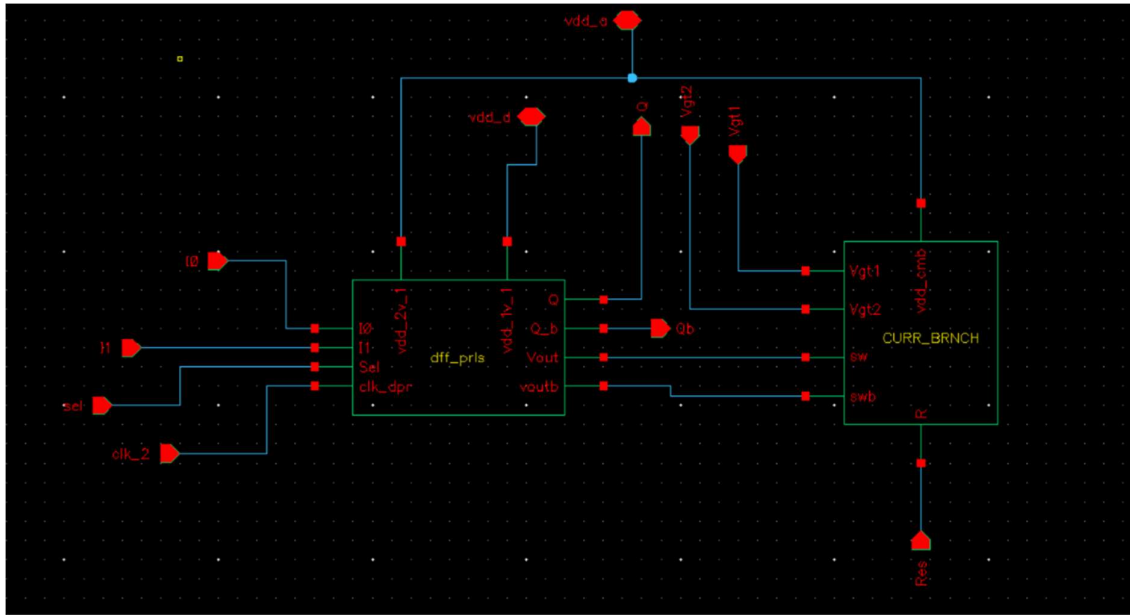


Fig:5 my **dff_prls** and **CURR_BRANCH** are connected together, so for every stage im providing the current through the current mirror connection (**current mirror comes in the last figure**) symbol name in cad – **Cmtm_dffls**

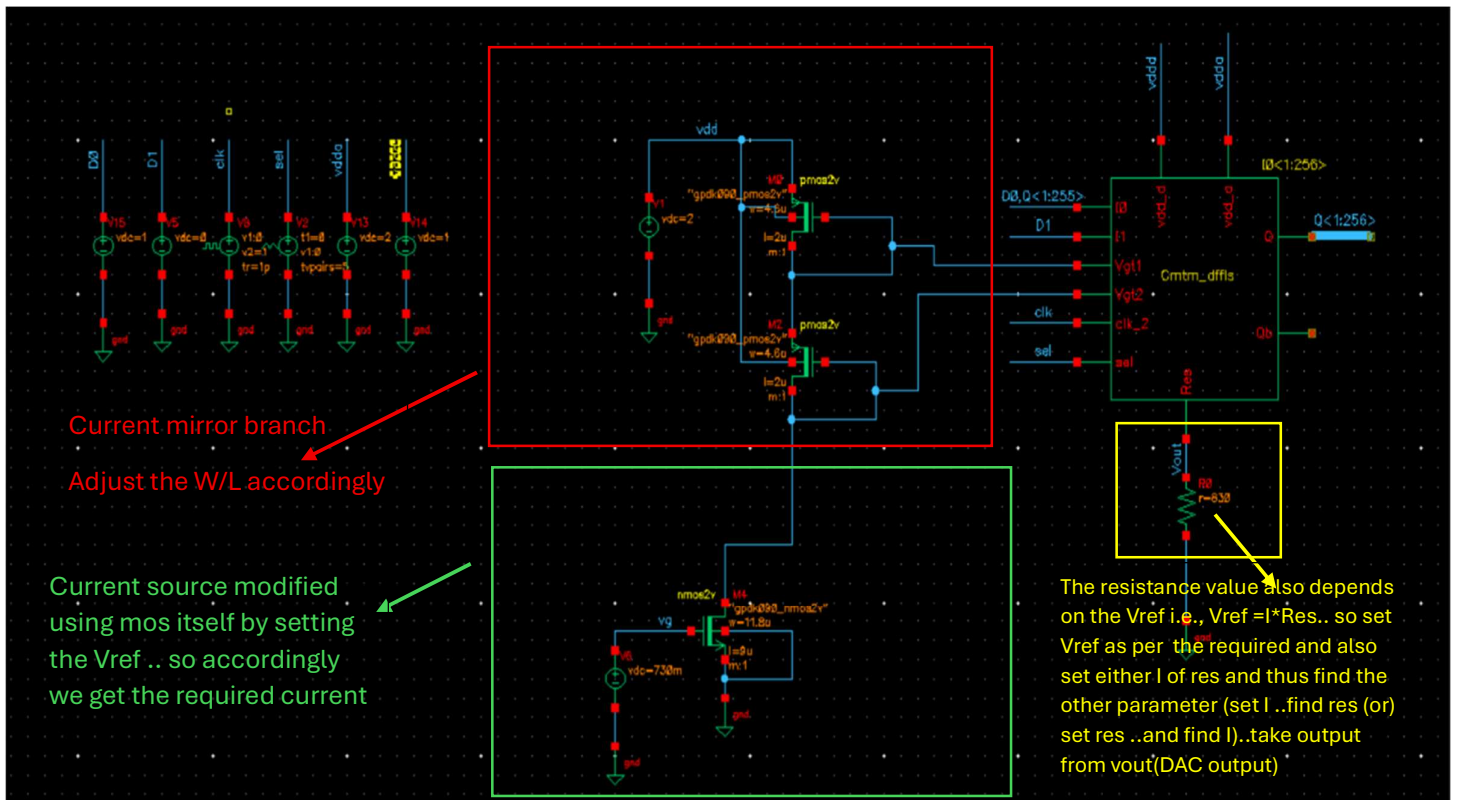
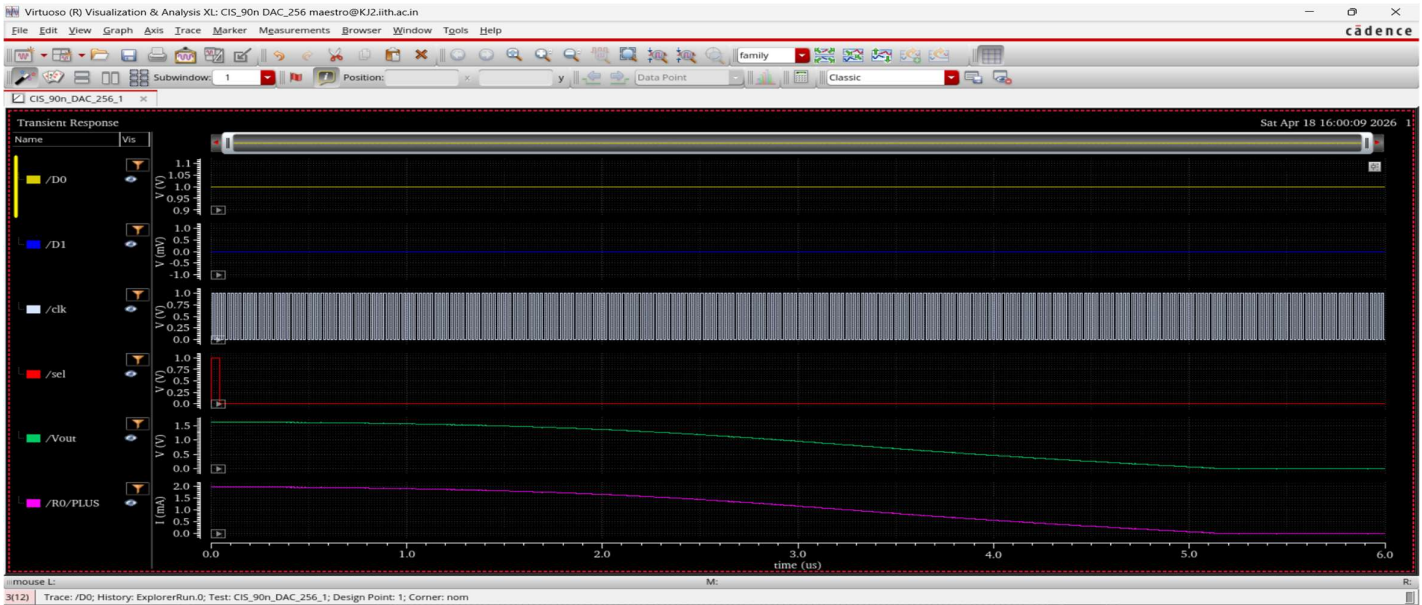


Fig:6 the complete connection of all blocks



Simulation result (Ro/plus if the vout node... highest voltage is 1.7v.. and the lowest is 6.64mV)

Example : I set my $V_{ref} = 1.7V$.

$\therefore$ I'm taking 256 levels $[2^8 = 8 \text{ bits}]$

$\therefore \frac{V_{ref}}{256 \text{ levels}} = \frac{1.7V}{256} = 6.64mV \approx 1LSB \text{ value}$

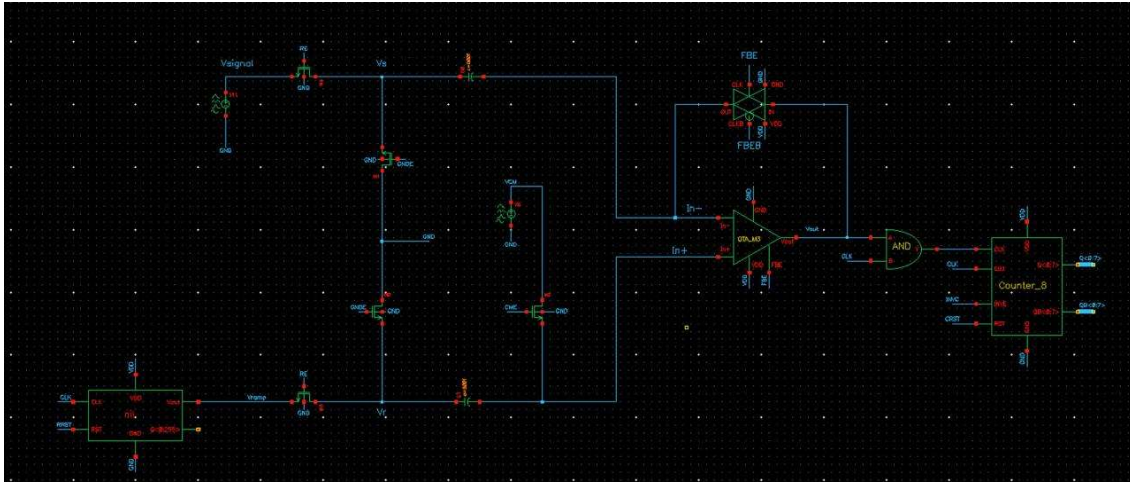
$\therefore V_{1LSB} = I \times R_L \Rightarrow$ I've set my I shld be as $8\mu A$.

$\therefore R_L = \frac{V_{1LSB}}{I} = \frac{6.64mV}{8\mu A} \approx 830\Omega$

$\therefore V_{ref} = 1.7V, V_{1LSB} = 6.64mV, I = 8\mu A, R_L = 830\Omega$

Now, Set w/L of MOSFETs [current mirror] to get that required current.

The overall circuitry for the column wise data acquisition of the image sensor is as follows:



The transmission gates were used to simply clear the offset that is been inside the op amp(comparator) so that the signal gets captured clearly. The counter depends on the clock and here used the digital CDS where the 2's compliment method is used to change the counter value while the execution.